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Smart Battery

Spacecraft Power and Data Handling - AE4901

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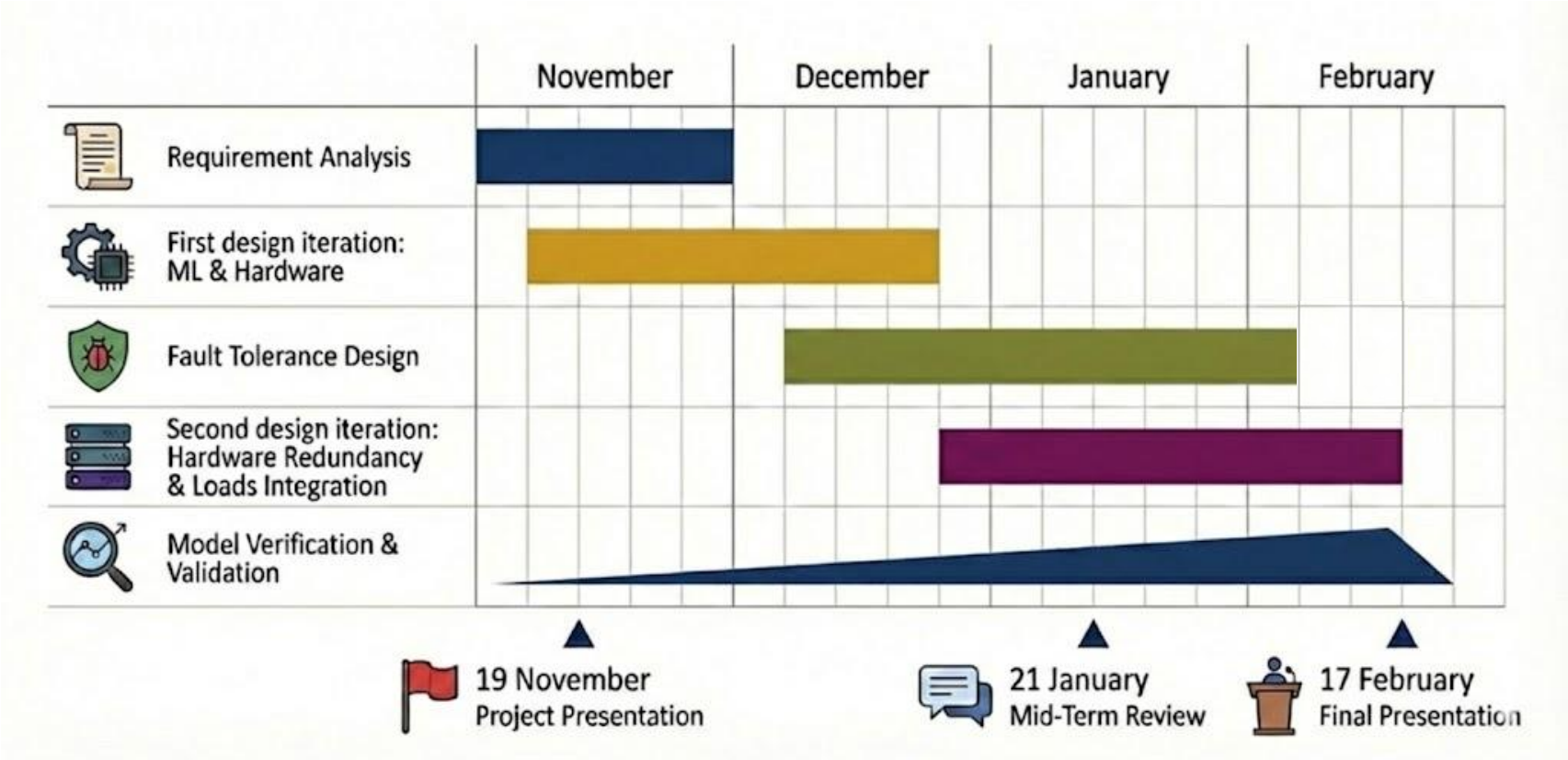
Demonstration

Future Improvements



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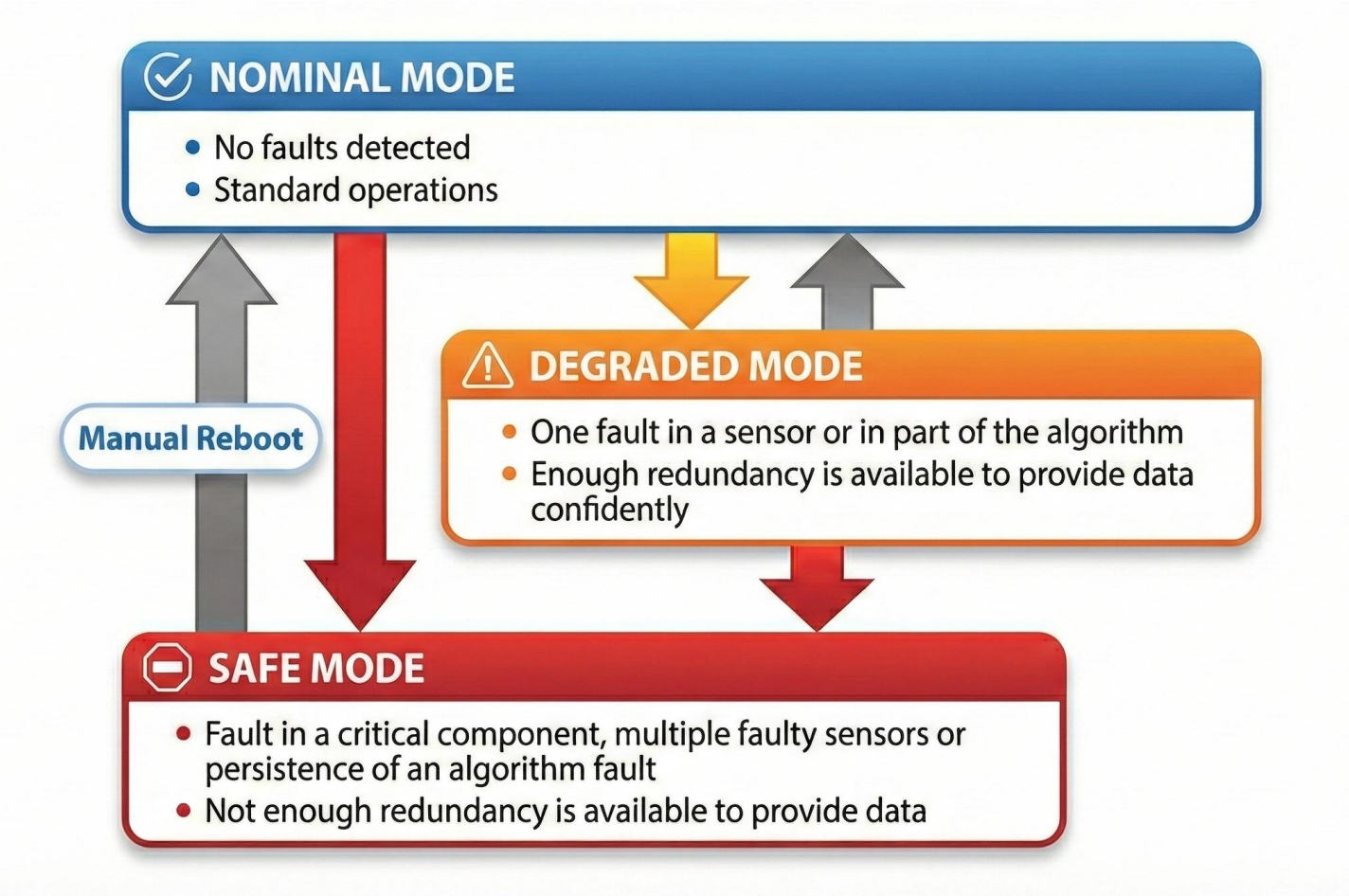
Project Timeline



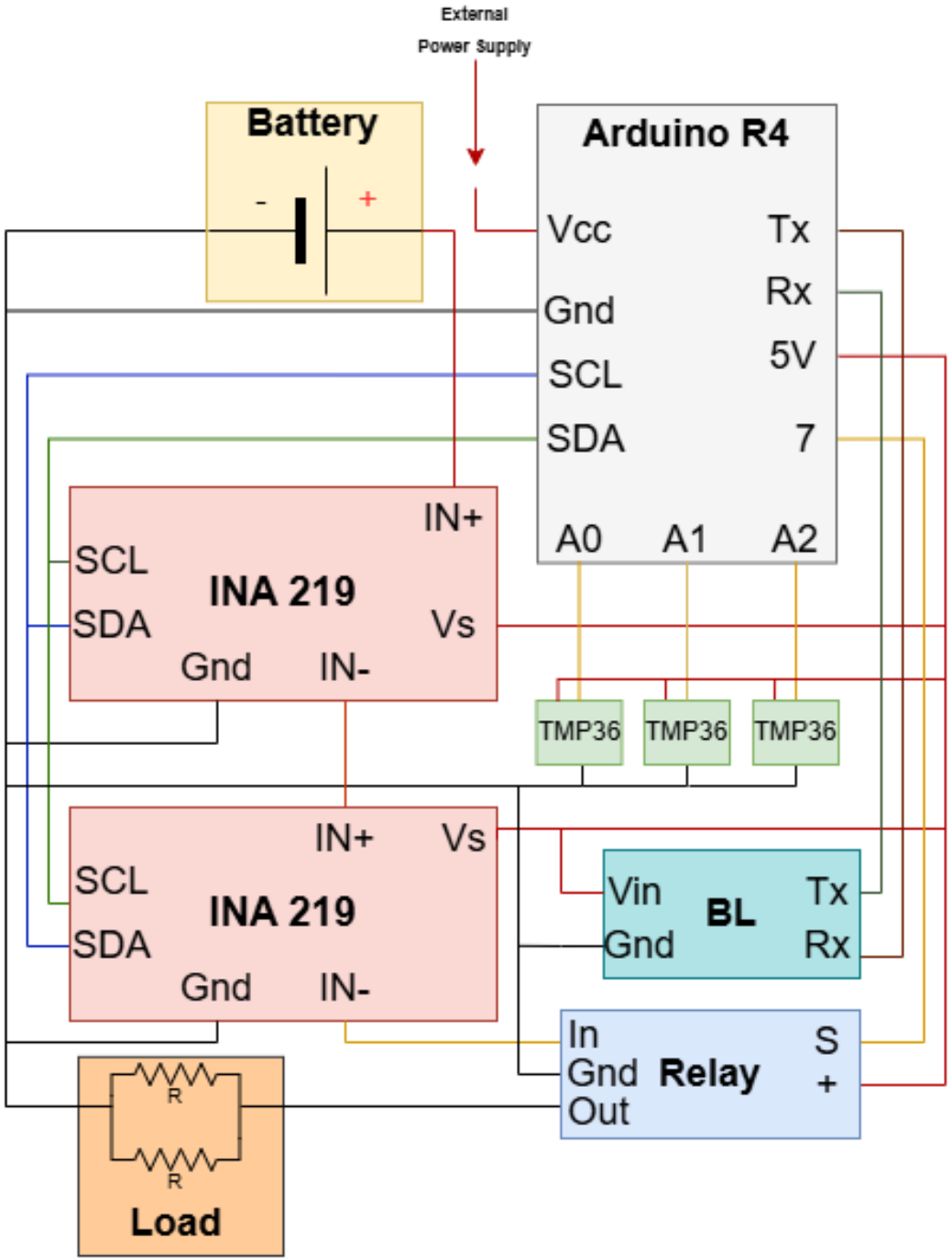
System Requirements

1. The system shall estimate State of Charge (SOC) with bounded error under varying operating conditions.
2. The system shall detect and react to current and temperature conditions exceeding predefined safety thresholds.
3. The system shall tolerate single faults without loss of core functionality and advise the user by entering a "degraded mode" in presence of partial failures.
4. The system shall not have any single point of failure apart from the Arduino microcontroller.

Operating Modes



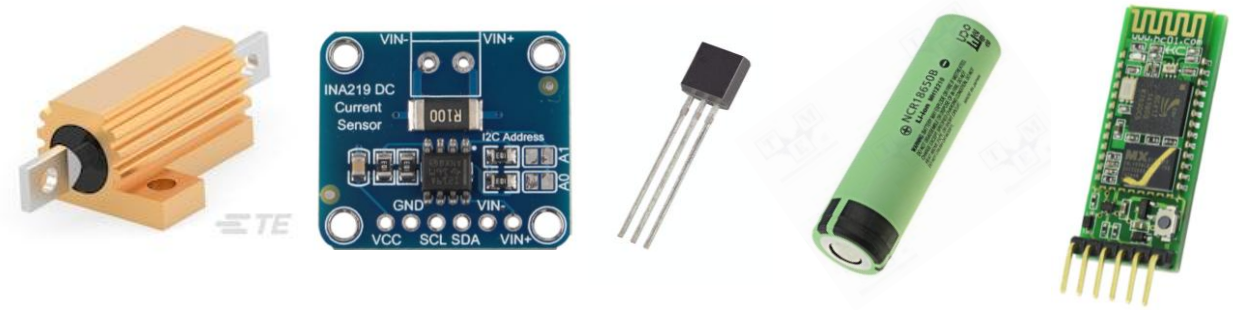
Hardware Architecture

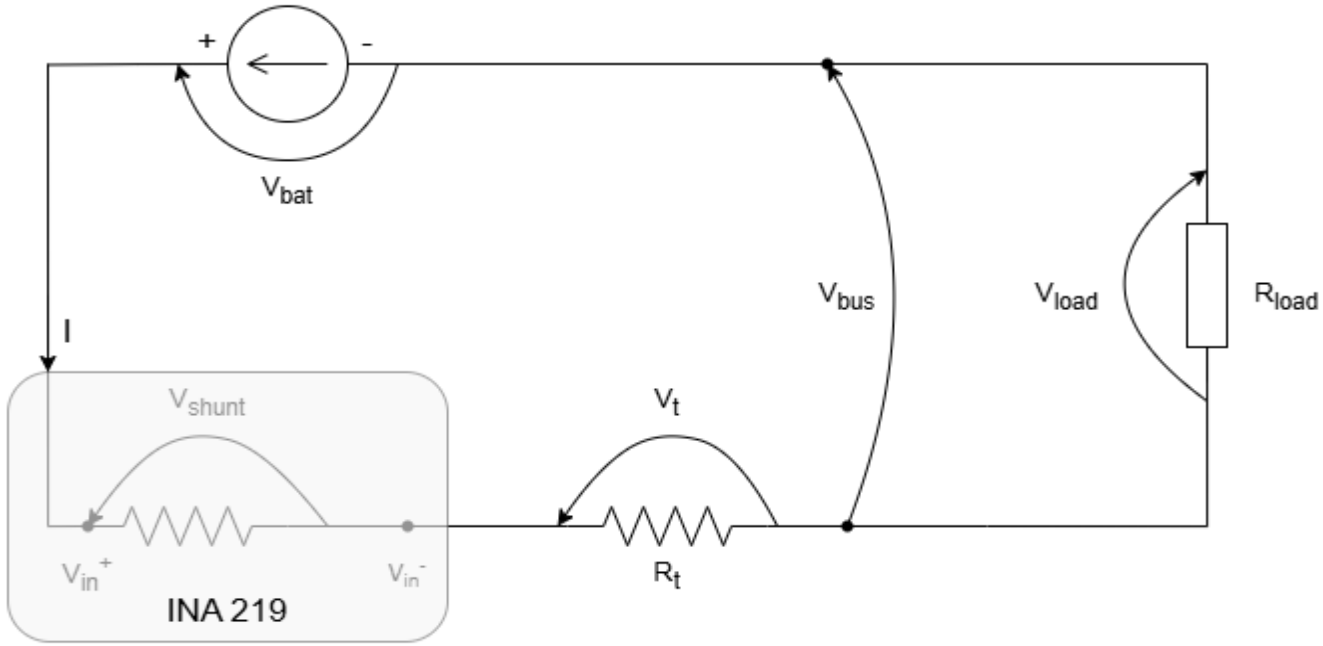


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Hardware Architecture

- 3 Temperature Sensors (TMP36GZ)
- 2 Current Sensors (INA219)
- Power Resistors (3.3Ω, 25W)
- Lithium-Ion Battery (Panasonic 18650B)
- Relay
- Bluetooth Module (HC-06)
- Wires, jumpers and breadboard

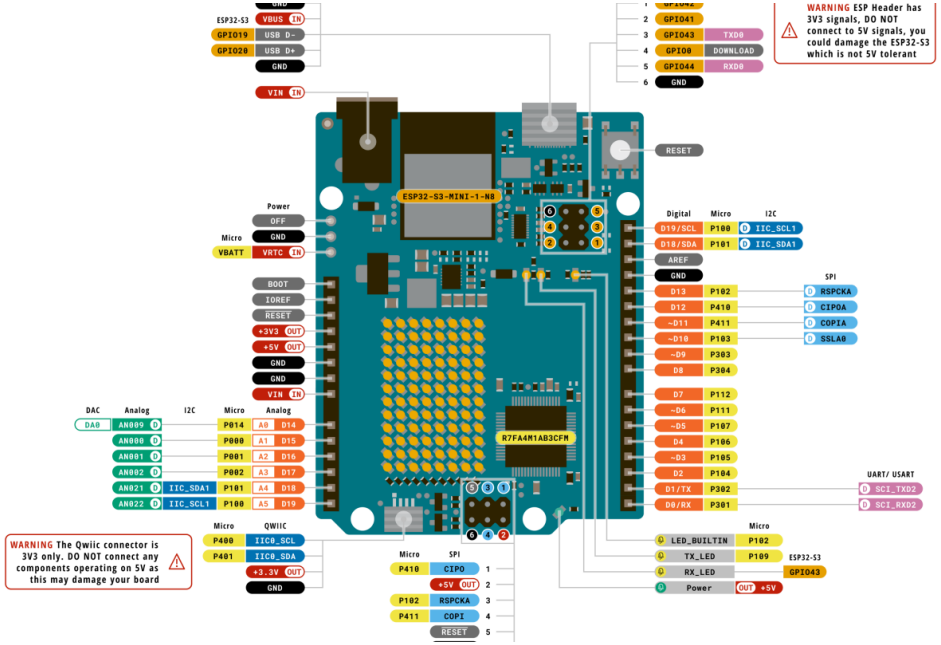


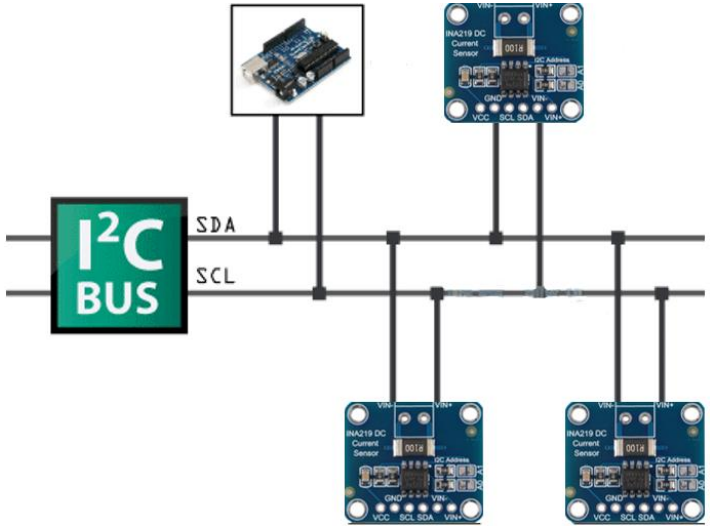


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Hardware Architecture

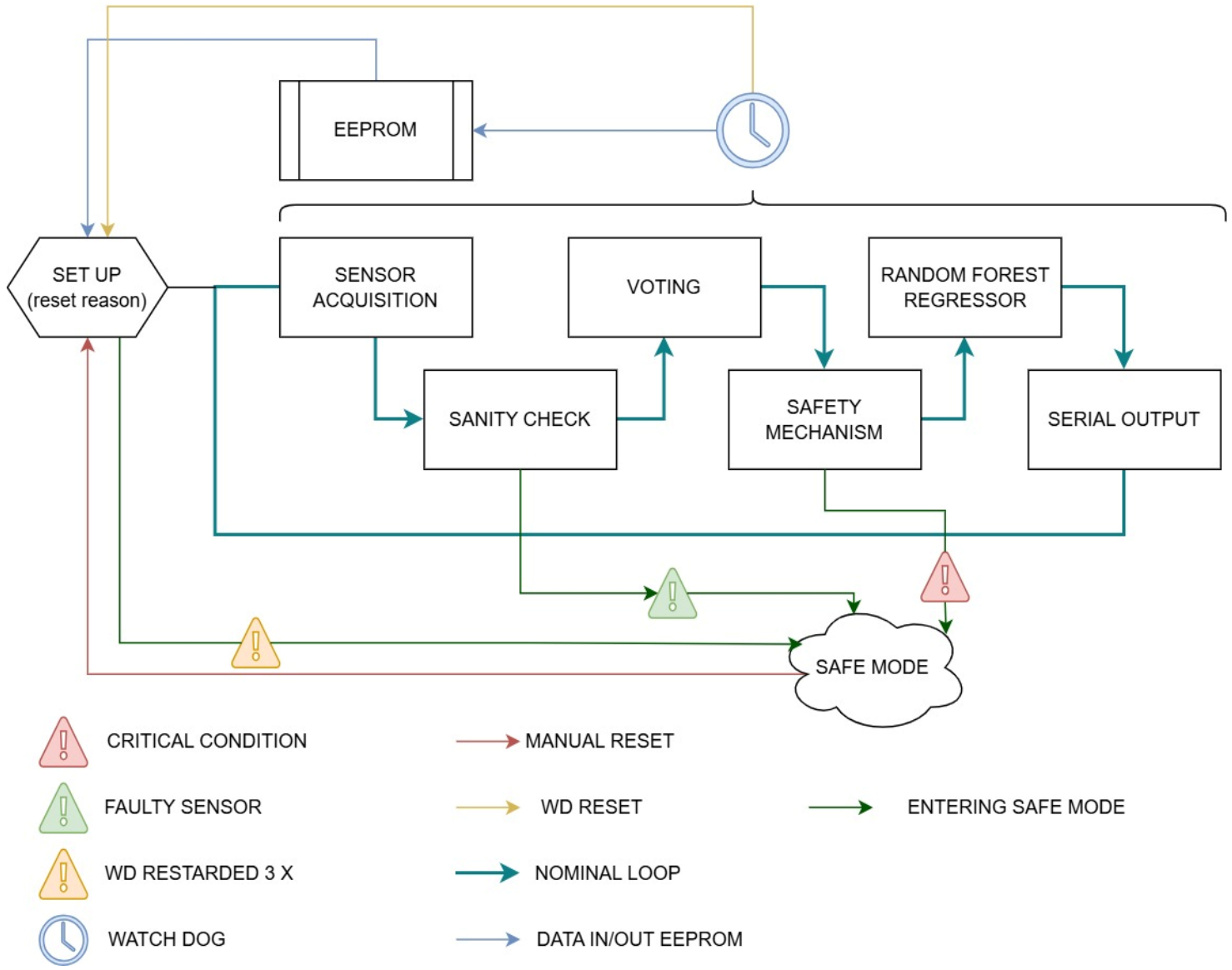
- Arduino R4:
  - 32-bit RA4M1 series microcontroller
  - 256 kB flash / 32 kB SRAM / 8 kB data flash – EEPROM
  - Real-time Clock
  - 14-bit ADC, up to 12-bit DAC
- Communication Protocols:
  - I2C (3 current sensors in parallel)
  - UART (Bluetooth Module)





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Software Architecture

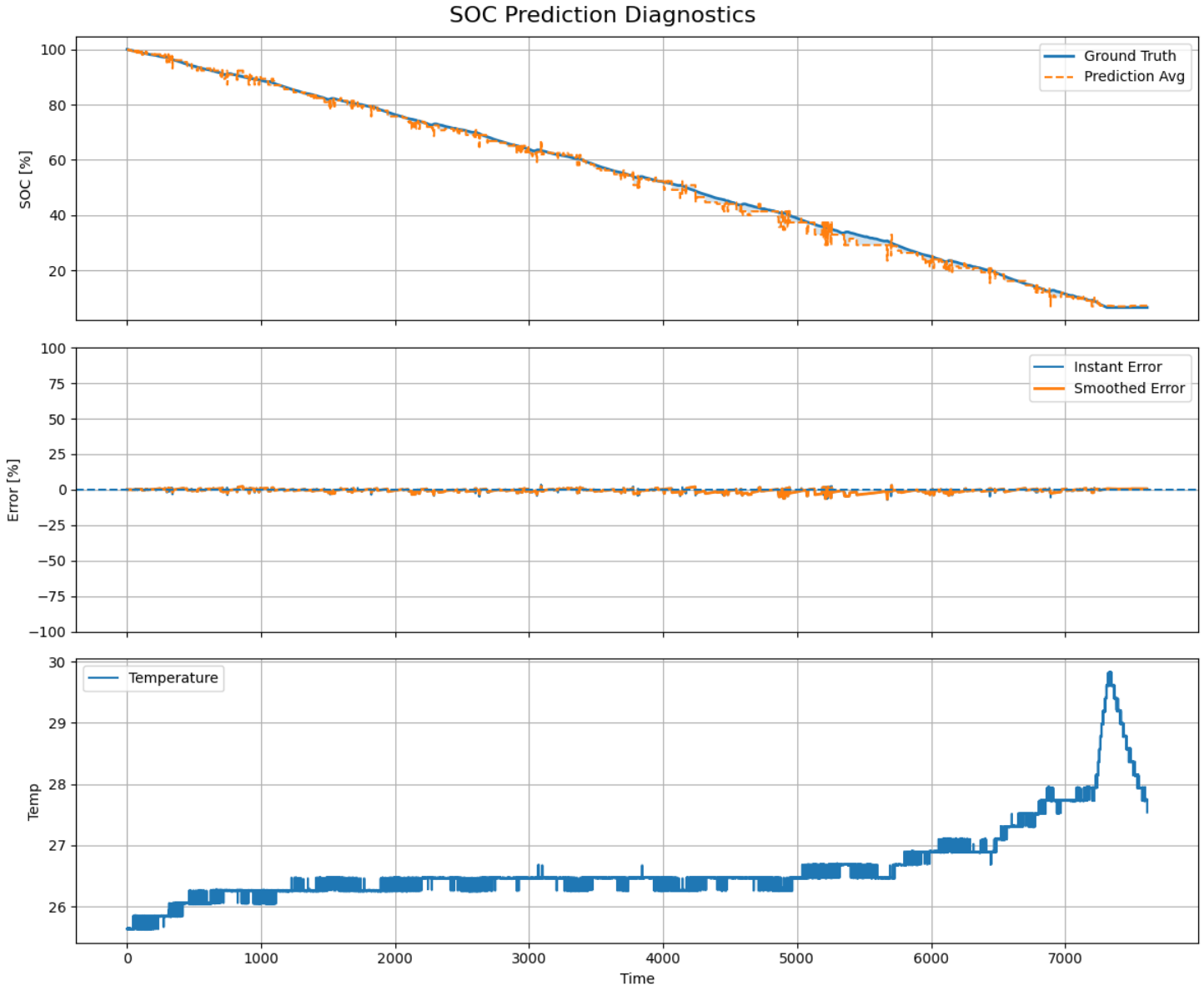


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Software Architecture

MACHINE LEARNING

- Panasonic Data-Set
  - +3mln data point
  - [-10°C, +25°C]
- Random Forest Regressor
- Down to only 54 kB algorithms

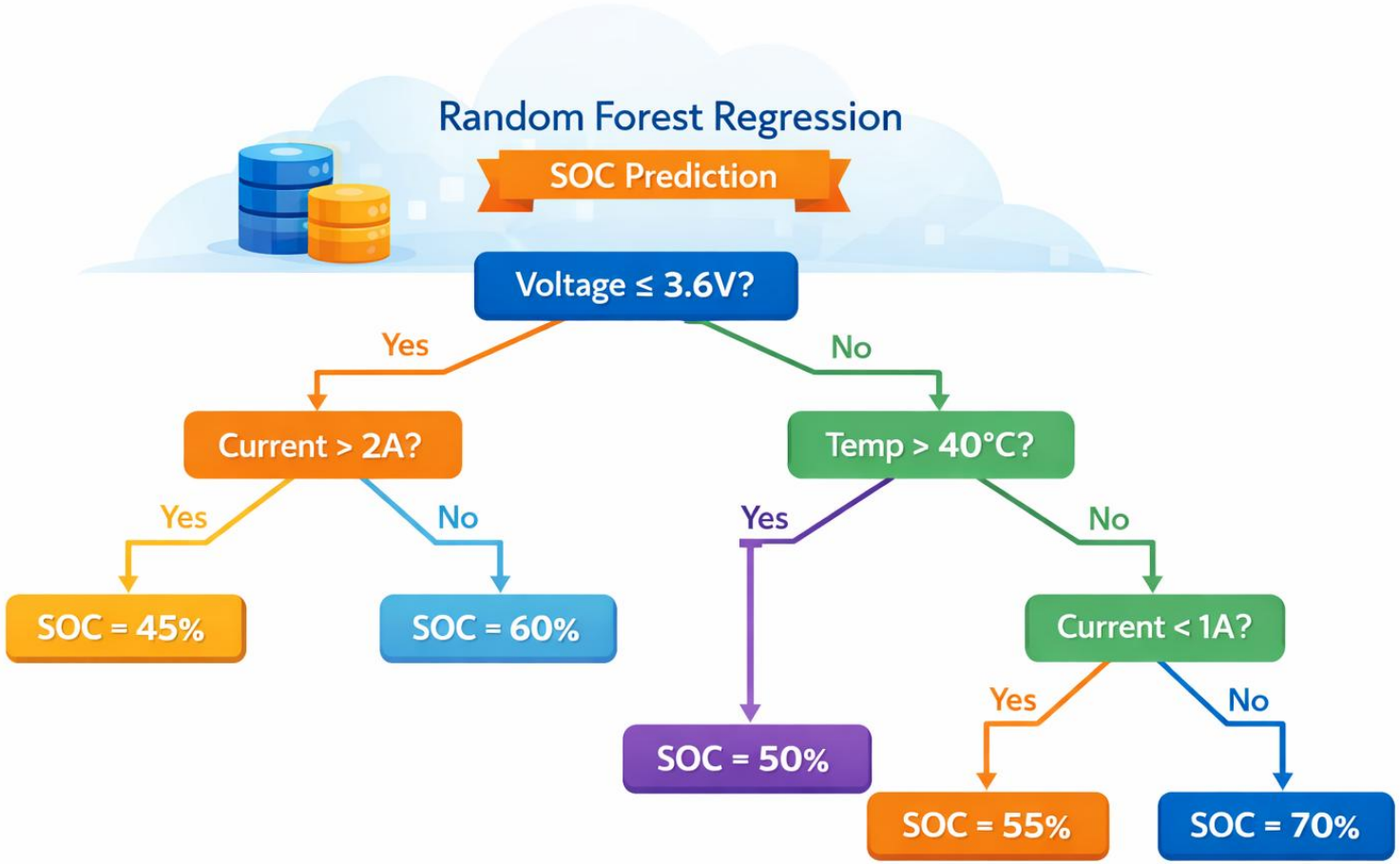


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Software Architecture

RANDOM FOREST

- Multiple trees estimating SOC (Redundancy)
- Simple working principle: threshold comparison
  - Easy translation to C/C++
  - Straightforward hardware implementation



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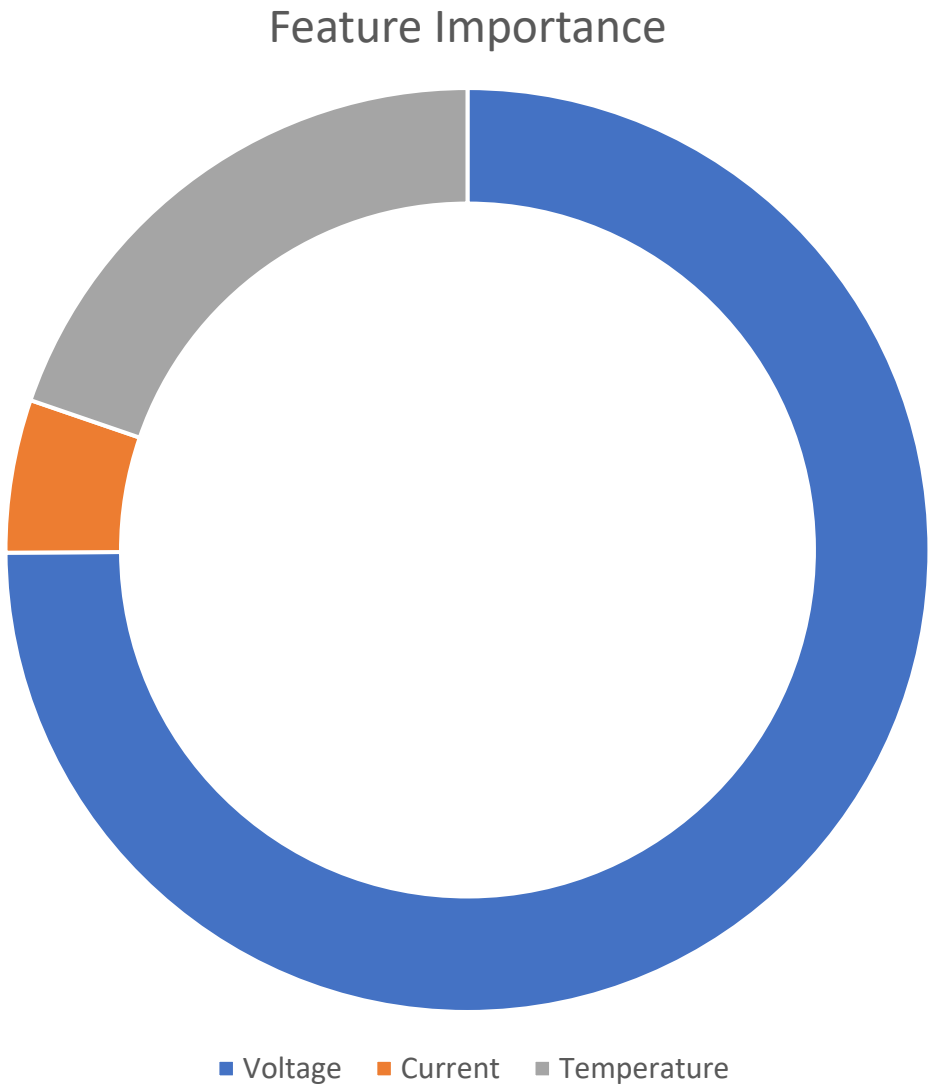
Software Architecture

RANDOM FOREST

Key Input Parameters:

- Voltage
- Temperature (Battery)
- Current

Importance proportional to how frequent within thresholds



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Fault Tolerance

SANITY CHECK

Fault Detection:

- Bounds
- Gradient
- Relative distances

Penalty system:

- Adding points when faulty
- Detracting on correct behavior

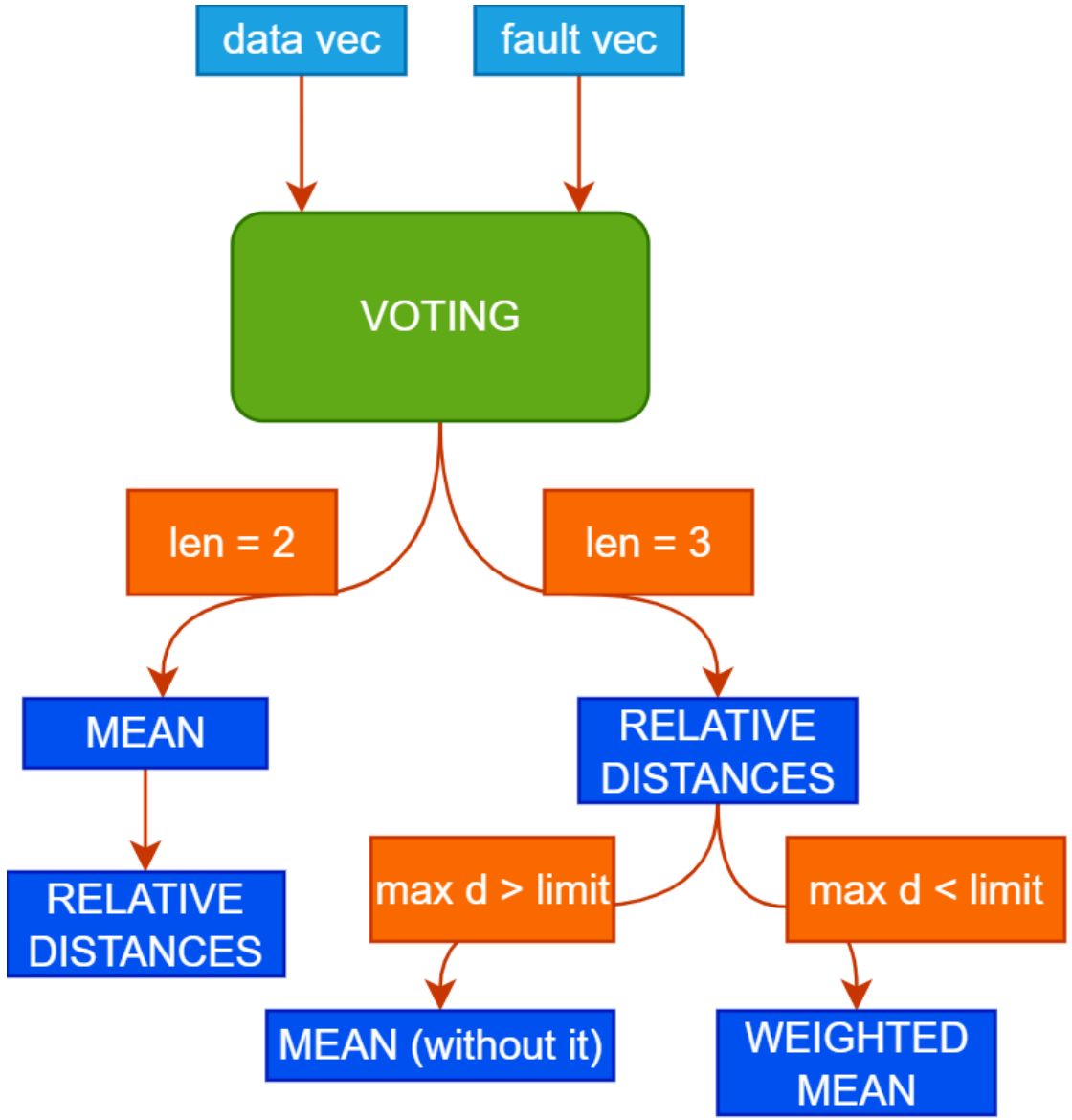


Fault Tolerance

$$\text{Mean: } res = \frac{v_0 + v_1}{2}$$

$$\text{Relative Distance: } d_i = \frac{\sum_{j=0, j\neq i}^{3} |v_j - v_i|}{\sum_{j=0}^{3} v_j}$$

$$\text{Weighted Mean: } res = \sum_{i=0}^{2} (1 - d_i) \cdot v_i$$



Fault Tolerance

SAFETY MECHANISM

- Keeping the battery distant from critic Temperatures and Currents
  - I < 5 A
  - T > 0°C (especially while charging)
  - T < 45°C
  - V > 2.7 V
- If not respected the system will enter the Safe Mode and cut-off the power



Fault Tolerance

WATCHDOG TIMER

Ensures continuous operation by automatically recovering from system hangs.

Timeout set to 0.4 s

Boot Loop Protection (EEPROM)

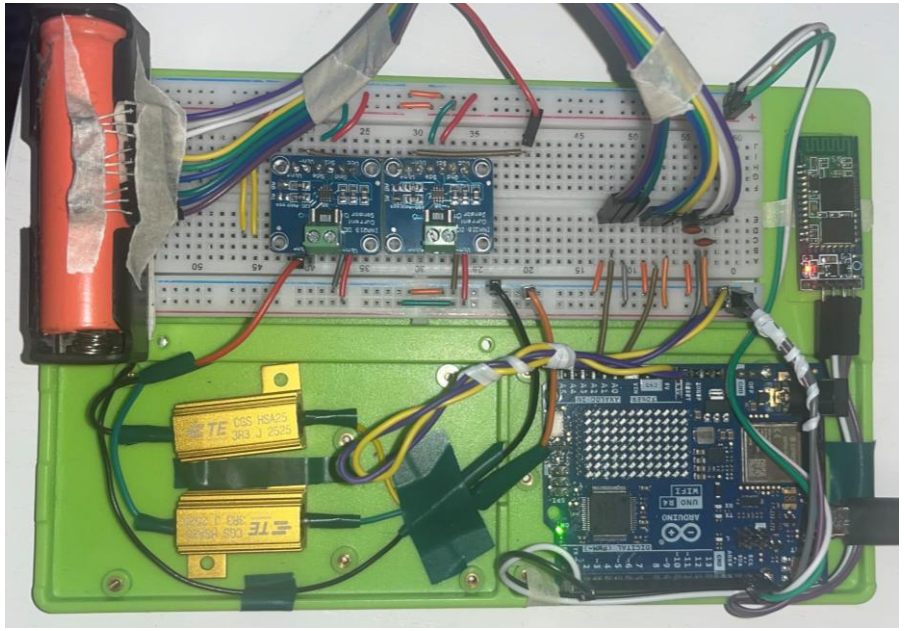
Safe Mode after 3 resets



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Testing Setup







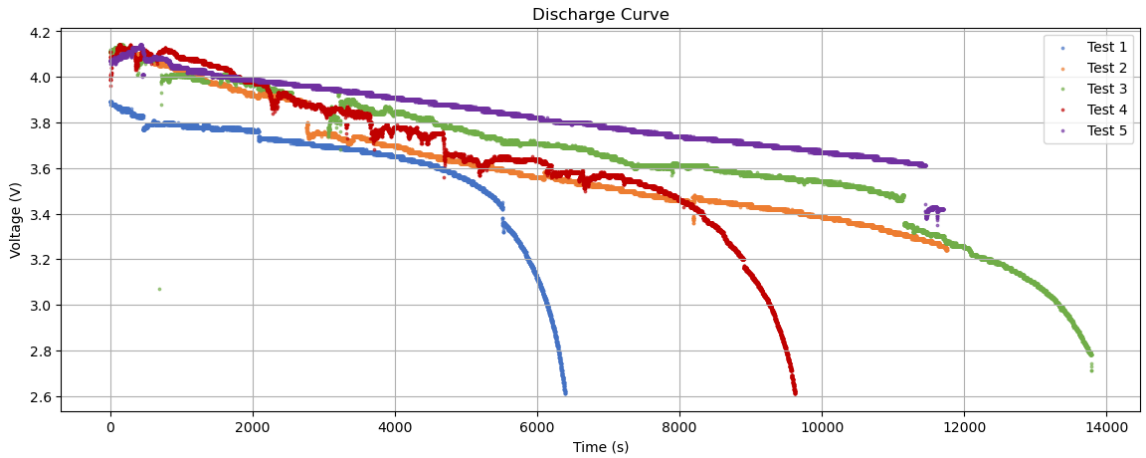
Charging

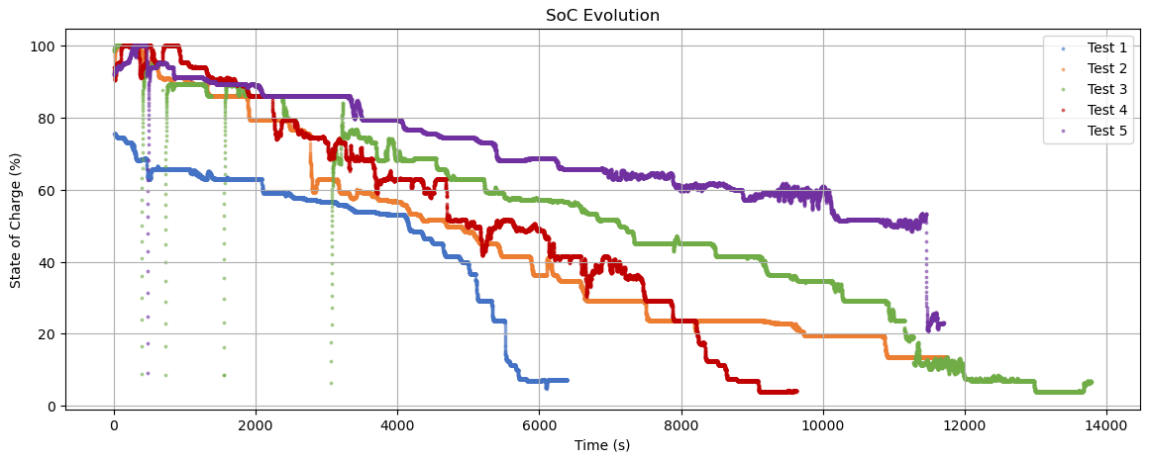
Discharging Test and Data Transmission

Data Collection

Model Verification

- Discharge at ~ 1 A
- 2 different batteries
- (Test 1, 3 & Tests 2, 4, 5)

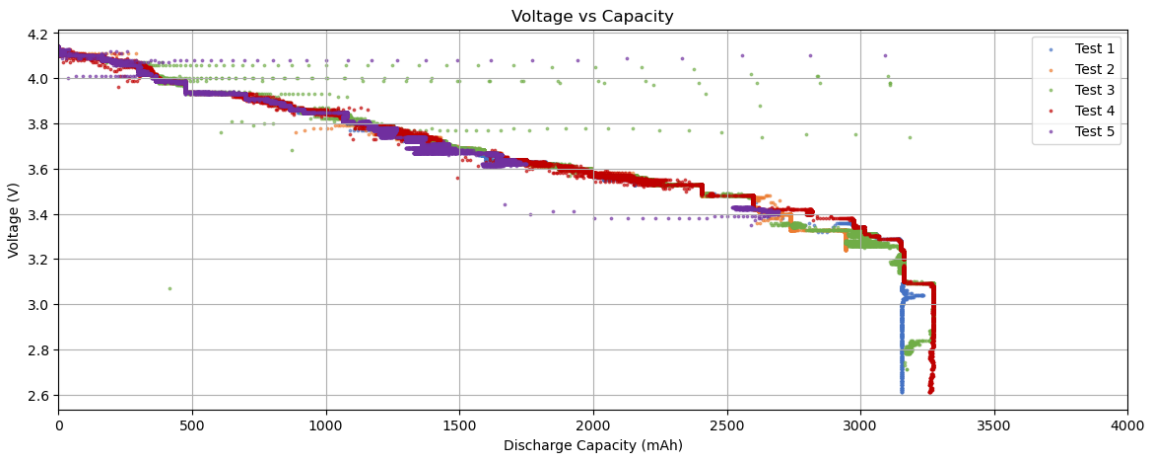


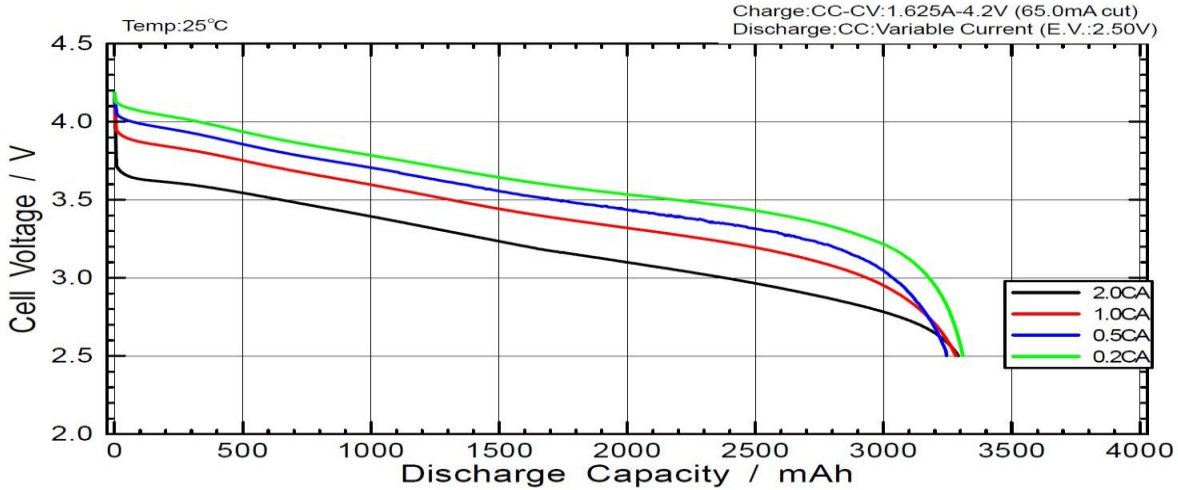


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Model Verification

- Verification with datasheet
- Strong correlation for both batteries





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Limitations & Future Improvements

LIMITATIONS



Single Point of Failure (Arduino)



Rough Impedance Measurements



No Predictions While Charging



Missing Sensor

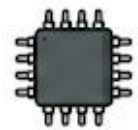


Affected Temp Measurement

FUTURE IMPROVEMENTS



Stand-by Pairing
(with boot memory)



Direct Soldering
(no breadboard, jumpers)



Comprehensive Datasets



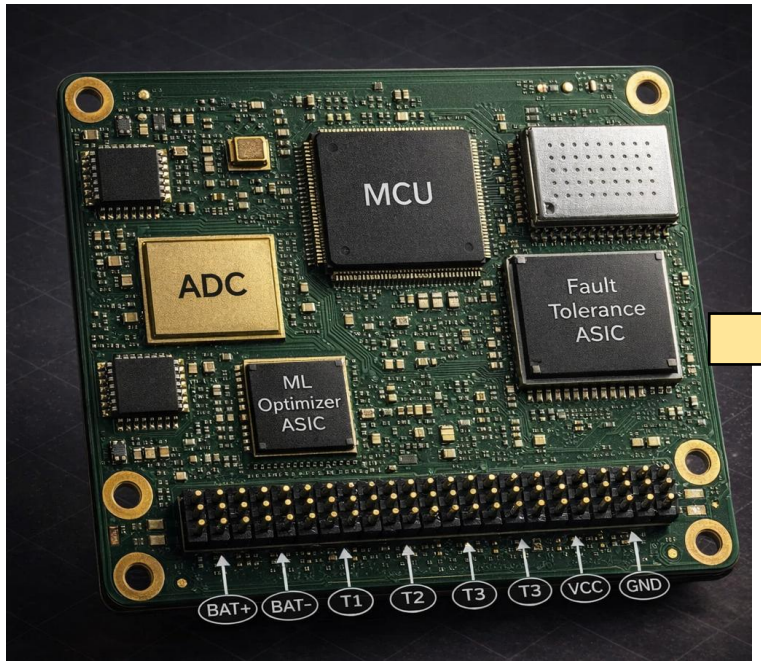
Add Third Sensor

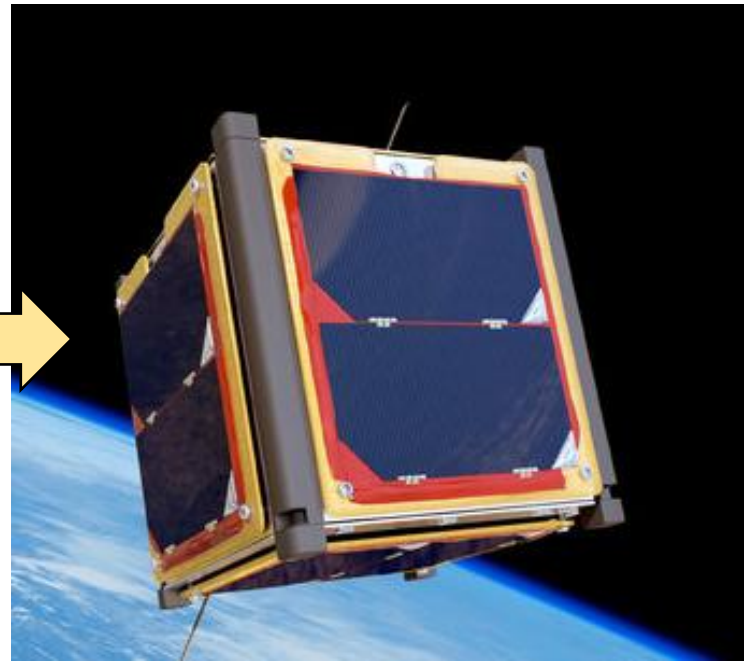


Better Thermal Contact
(between sensor & battery surface)

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Final Product Concept





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